

Hoja de Trabajo 2

Verifique las identidades

$$1) \frac{1}{1 + \sec t} = (\sec t - \tan t) \sec t$$

$$\frac{1}{1 + \sec t} = \left(\frac{1}{\cos t} - \frac{\sec t}{\cos t} \right) \frac{1}{\cos t}$$

$$\frac{1}{1 + \sec t} = \left(\frac{1 - \sec t}{\cos t} \right) \frac{1}{\cos t}$$

$$\frac{1}{1 + \sec t} = \frac{1 - \sec t}{\cos^2 t}$$

$$\frac{1}{1 + \sec t} = \frac{1 - \sec t}{1 - \sec^2 t}$$

$$\frac{1}{1 + \sec t} = \frac{1 - \sec t}{(1 - \sec t)(1 + \sec t)}$$

$$\frac{1}{1 + \sec t} = \frac{1}{1 + \sec t}$$

$$2) \tan 2\theta = \frac{2 \cos \theta}{\csc^2 \theta - 2}$$

$$\frac{2 \tan 2\theta}{1 - \tan^2 2\theta} = \frac{2 \cos \theta}{\frac{1}{\sin^2 \theta} - 2}$$

$$\frac{2 \frac{\sin \theta}{\cos \theta}}{1 - \frac{\sin^2 \theta}{\cos^2 \theta}} = \frac{2 \cos \theta}{\frac{1}{\sin^2 \theta} - 2}$$

$$\frac{2 \operatorname{sen} \theta}{\cos \theta} = \frac{2 \cos \theta \operatorname{sen}^2 \theta}{1 - 2 \operatorname{sen}^2 \theta}$$

No es una identidad

$$\frac{2 \operatorname{sen} \theta \cos^2 \theta}{\cos 2\theta} = \frac{\operatorname{sen} 2\theta (\operatorname{sen} \theta)}{1 - 2 \operatorname{sen}^2 \theta}$$

$$3) \frac{1 - \operatorname{sen} a}{\sqrt{1 + \operatorname{sen} a}} = \frac{|\cos a|}{1 + \operatorname{sen} a}$$

$$\frac{\sqrt{1 - \operatorname{sen} a}}{\sqrt{1 + \operatorname{sen} a}} = \frac{|\cos a|}{1 + \operatorname{sen} a}$$

$$\frac{\sqrt{1 - \operatorname{sen} a}}{\sqrt{1 + \operatorname{sen} a}} \neq \frac{\sqrt{1 + \operatorname{sen} x}}{\sqrt{1 + \operatorname{sen} x}} = \frac{\sqrt{(1 - \operatorname{sen} a)(1 + \operatorname{sen} a)}}{\sqrt{1 + \operatorname{sen} a} \sqrt{1 + \operatorname{sen} a}} = \frac{\sqrt{1 - \operatorname{sen}^2 a}}{1 + \operatorname{sen} a}$$

$$\frac{\sqrt{\cos^2 a}}{1 + \operatorname{sen} a} = \frac{\cos a}{1 + \operatorname{sen} a} = \frac{\cos a}{1 + \operatorname{sen} a}$$

$$4) \tan \frac{1}{2} u = \csc u - \cot u$$

$$\tan \frac{1}{2} u = \frac{1}{\operatorname{sen} u} - \frac{\cos u}{\operatorname{sen} u}$$

$$\tan \frac{1}{2} u = \frac{1 - \cos u}{\operatorname{sen} u}$$

$$\tan \frac{1}{2} u = \tan \frac{1}{2} u$$

Encuentre todas las soluciones de la ecuación

$$1) 2\cos x + \tan x = \sec x$$

$$2\cos x + \frac{\sin x}{\cos x} = \frac{1}{\cos x}$$

$$2\cos x + \frac{\sin x}{\cos x} - \frac{1}{\cos x} = 0$$

$$\frac{2\cos^2 x + \sin x - 1}{\cos x} = 0$$

$$2\cos^2 x + \sin x - 1 = 0$$

$$(2\cos^2 x - 1) + \sin x = 0$$

$$\cos 2x + \sin x = 0$$

$$1 - 2\sin^2 x + \sin x = 0$$

$$2\sin^2 x - \sin x - 1 = 0$$

$$(2\sin x + 1)(\sin x - 1) = 0$$

$$2\sin x + 1 = 0$$

$$\sin x = -1/2$$

$$x = \sin^{-1}(-1/2)$$

$$\sin x - 1 = 0$$

$$\sin x = 1$$

$$x = \sin^{-1} 1$$

$$x = \frac{7\pi}{6} + 2\pi n, \frac{11\pi}{6} + 2\pi n$$

$$x = \frac{\pi}{2}$$

$$2) \operatorname{sen} x + 2 \cos^2 x = 1$$

$$\operatorname{sen} x + 2(1 - \operatorname{sen}^2 x) - 1 = 0$$

$$\operatorname{sen} x + 2 - 2 \operatorname{sen}^2 x - 1 = 0$$

$$\operatorname{sen} x + 1 - 2 \operatorname{sen}^2 x = 0$$

$$-2 \operatorname{sen}^2 x + \operatorname{sen} x + 1 = 0$$

$$-(\operatorname{sen} x - 1)(2 \operatorname{sen} x + 1) = 0$$

$$\operatorname{sen} x - 1 = 0$$

$$\operatorname{sen} x = 1$$

$$x = \operatorname{sen}^{-1} 1$$

$$2 \operatorname{sen} x + 1 = 0$$

$$\operatorname{sen} x = -1/2$$

$$x = \operatorname{sen}^{-1} -1/2$$

$$x = \frac{\pi}{2} + 2\pi n$$

$$x = \frac{7\pi}{6} + 2\pi n, \frac{11\pi}{6} + 2\pi n$$

Encuentra las soluciones de la ecuación en el intervalo de 0 a 2π

$$1) \operatorname{sen} x \cos 2x + \cos x \operatorname{sen} 2x = 0$$

$$\operatorname{sen}(x + 2x) = 0$$

$$\operatorname{sen} 3x = 0$$

$$3x = \operatorname{sen}^{-1} 0$$

$$3x = 0, \pi$$

$$x = \frac{\pi}{3}$$

$$2) \sin 5x = \sin 3x$$

$$\sin 5x - \sin 3x = 0$$

$$2 \cos \frac{5x+3x}{2} \sin \frac{5x-3x}{2} = 0$$

$$2 \cos 4x \sin x = 0$$

$$\cos 4x \sin x = 0$$

$$\cos 4x = 0$$

$$4x = \cos^{-1} 0$$

$$\sin x = 0$$

$$x = \sin^{-1} 0$$

$$4x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$x = 0, \pi$$

$$x = \pi$$

$$x = \frac{\pi}{2} \div 4 = \frac{\pi}{8} + \frac{\pi n}{4}$$

$$x = \frac{3\pi}{2} \div 4 = \frac{3\pi}{8}$$

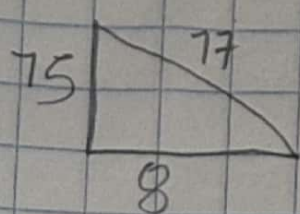
Encuentre el valor exacto de la expresión siempre que esté definido.

$$1) \arccos(\tan \frac{3\pi}{4})$$

$$\arccos(-1) = \pi$$

$$2) \cos\left(\sin^{-1}\frac{15}{17} - \sin^{-1}\frac{8}{17}\right)$$

$$\cos\left(\sin^{-1}\left(\frac{15}{17}\right)\right) \cos\left(\sin^{-1}\frac{8}{17}\right) + \sin\left(\sin^{-1}\frac{15}{17}\right) \sin\left(\sin^{-1}\frac{8}{17}\right)$$



$$\left(\frac{8}{17}\right)\left(\frac{15}{17}\right) + \left(\frac{15}{17}\right)\left(\frac{8}{17}\right) = \frac{240}{289}$$

